

Introduction to Financial Statement Analysis

- **Definition:** Financial statement analysis is the process of reviewing and evaluating a company's financial statements to gain insights into its financial performance, position, and cash flows.
- **Purpose:** To assess a company's past performance, current financial health, and future potential. Used by investors, creditors, management, and regulators.
- **Key Financial Statements:**
 - **Balance Sheet:** Snapshot of assets, liabilities, and owner's equity at a specific point in time.
 - **Income Statement (Profit & Loss Account):** Shows revenues, expenses, and profit/loss over a period.
 - **Cash Flow Statement:** Details cash inflows and outflows from operating, investing, and financing activities over a period.

Objectives and Importance

- **Objectives:**
 - Assess profitability (e.g., net profit margin, ROI).
 - Evaluate solvency (ability to meet long-term obligations).
 - Measure liquidity (ability to meet short-term obligations).
 - Analyze efficiency (how effectively assets are used).
 - Facilitate forecasting and planning.
 - Assist in decision-making (investment, lending, management).
- **Importance:**
 - **Investors:** To make informed investment decisions.
 - **Creditors/Lenders:** To assess creditworthiness and repayment capacity.
 - **Management:** For internal decision-making, performance evaluation, and strategic planning.
 - **Government/Regulators:** For taxation, economic policy, and compliance.

Tools and Techniques of Analysis

- **1. Comparative Statements:**
 - Compare financial data over two or more periods (e.g., year-on-year).
 - Show absolute changes and percentage changes.
 - Help identify trends and significant shifts in financial performance.
 - Example: Comparing sales revenue, cost of goods sold, and net profit over 3 years.
- **2. Common-Size Statements:**
 - Express each item in a financial statement as a percentage of a base figure.
 - **Income Statement:** Each item as a percentage of Net Sales.
 - **Balance Sheet:** Each item as a percentage of Total Assets.
 - Facilitate comparison between companies of different sizes or over different periods.
 - Example: Inventory as % of Total Assets; Gross Profit as % of Net Sales.
- **3. Ratio Analysis:**
 - Calculates relationships between different financial statement items.
 - Provides insights into liquidity, solvency, profitability, and efficiency.
 - **Categories of Ratios:**
 - Liquidity Ratios
 - Solvency Ratios
 - Profitability Ratios
 - Activity/Efficiency Ratios
 - Market Ratios (though less relevant for FMM Class 12, sometimes included)

Liquidity Ratios

- Measure a company's ability to meet its short-term obligations.
- **1. Current Ratio:**
 - Formula: $\text{Current Assets} / \text{Current Liabilities}$
 - Ideal Ratio: 2:1 (generally, but varies by industry)
 - Indicates: Short-term debt-paying ability.
- **2. Quick Ratio (Acid-Test Ratio):**
 - Formula: $(\text{Current Assets} - \text{Inventory}) / \text{Current Liabilities}$
 - Ideal Ratio: 1:1 (generally)
 - Indicates: Ability to meet short-term obligations without relying on inventory sales.

Solvency Ratios

- Measure a company's ability to meet its long-term obligations.

- **1. Debt-to-Equity Ratio:**
 - Formula: $\text{Total Debt} / \text{Shareholders' Equity}$
 - Indicates: Proportion of debt financing relative to equity financing. A high ratio implies higher risk.
- **2. Debt-to-Assets Ratio:**
 - Formula: $\text{Total Debt} / \text{Total Assets}$
 - Indicates: Proportion of assets financed by debt.
- **3. Interest Coverage Ratio:**
 - Formula: $\frac{\text{EBIT (Earnings Before Interest \& Taxes)}}{\text{Interest Expense}}$
 - Indicates: Company's ability to meet its interest payments. A higher ratio is better.

Profitability Ratios

- Measure a company's ability to generate profit from its sales, assets, or equity.
- **1. Gross Profit Ratio:**
 - Formula: $(\text{Gross Profit} / \text{Net Sales}) \times 100$
 - Indicates: Profitability of sales after deducting cost of goods sold.
- **2. Net Profit Ratio:**
 - Formula: $(\text{Net Profit after Tax} / \text{Net Sales}) \times 100$
 - Indicates: Overall efficiency of the business, including operating and non-operating activities.
- **3. Return on Investment (ROI) / Return on Capital Employed (ROCE):**
 - Formula: $(\text{EBIT} / \text{Capital Employed}) \times 100$
 - $\text{Capital Employed} = \text{Shareholders' Funds} + \text{Long-term Debt OR Total Assets} - \text{Current Liabilities}$.
 - Indicates: Efficiency of using capital to generate profits.
- **4. Earnings Per Share (EPS):**
 - Formula: $(\text{Net Profit after Tax} - \text{Preference Dividends}) / \text{Number of Equity Shares}$
 - Indicates: Profit attributable to each equity share.

Activity/Efficiency Ratios

- Measure how efficiently a company uses its assets to generate sales.
- **1. Inventory Turnover Ratio:**
 - Formula: $\text{Cost of Revenue from Operations} / \text{Average Inventory}$
 - Indicates: How quickly inventory is sold and replaced. A higher ratio is generally better.
- **2. Trade Receivables Turnover Ratio (Debtors Turnover Ratio):**
 - Formula: $\text{Net Credit Revenue from Operations} / \text{Average Trade Receivables}$
 - Indicates: Efficiency of collecting debts from customers.
- **3. Trade Payables Turnover Ratio (Creditors Turnover Ratio):**
 - Formula: $\text{Net Credit Purchases} / \text{Average Trade Payables}$
 - Indicates: How quickly the company pays its suppliers.
- **4. Working Capital Turnover Ratio:**
 - Formula: $\text{Revenue from Operations} / \text{Working Capital}$
 - $\text{Working Capital} = \text{Current Assets} - \text{Current Liabilities}$
 - Indicates: Efficiency of using working capital to generate sales.

Limitations of Financial Statement Analysis

- **1. Historical Data:** Based on past performance, which may not be indicative of future results.
- **2. Accounting Policies:** Different accounting methods (e.g., inventory valuation, depreciation) can distort comparisons.
- **3. Inflationary Effects:** Financial statements do not fully account for changes in purchasing power due to inflation.
- **4. Qualitative Factors Ignored:** Non-financial aspects like management quality, brand reputation, labor relations, etc., are not reflected.
- **5. Window Dressing:** Companies may manipulate financial statements to present a better picture.
- **6. Industry Differences:** Comparisons between companies in different industries can be misleading due to varying norms.
- **7. Lack of Standardized Definitions:** Ratios can be calculated differently, making comparisons difficult.